

Wave 5.2R Integrated Specialist Model Campaign Results

Wave 5.2R Empirical Specialist Closeout

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Summary

The approved empirical Wave 5.2R campaign completed all 24 / 24 planned entries with no failed queue item: three frozen replay controls, eighteen single-branch training runs, and three conditional A08 runs. All 21 trained run directories contain the required checkpoint, training history, test predictions, and metrics summary.

Only the forward-gated centered H08 branch (A02) passed its predeclared specialty and multi-index non-regression gate. It passed for all three seeds, so A08 executed with A02 as its only admitted branch and reproduced the same predictions. The campaign outcome is therefore forward_harmonic_specialist_added , not a multi-specialist qualification.

A04 , seed 271828 , is the validation-only scalar winner with global validation MAE 0.001539071 deg and global test MAE 0.001456655 deg . However, A04 failed both its centered-shape specialty gate and the multi-index non-regression gate. It is recorded as a provisional family winner only. No accepted model, incumbent, program registry, or deployment status changes in this closeout.

Campaign Contract

- Campaign: wave52r_integrated_specialist_model_2026_08_02 .
- Execution: LAN-remote, CUDA-enabled campaign runner.
- Dataset and inputs: polished_dataset + setpoints .
- Surfaces: separate Fw , Bw , and direction-aware global .
- Seeds: 314159 , 271828 , and 161803 .
- Started: 2026-08-03T17:49:23+02:00 .
- Completed: 2026-08-03T17:58:38+02:00 .
- Runtime target-derived input count: 0 for every trained run.
- Selection boundary: validation-only scalar ordering plus predeclared branch-specific gates; no automatic promotion. This remains an empirical integrated-specialist study. It is not a PINN, does not reopen physics-integrated Wave 6, and does not establish TwinCAT or TF3820 runtime readiness.

Completion And Integrity

Evidence	Expected	Observed	Result
Campaign entries	24	24	Pass
Frozen replay controls	3	3	Pass
Trained run directories	21	21	Pass
Queue-state files	24	24	Pass

Evidence	Expected	Observed	Result
Required campaign artifacts	8	8	Pass
Required trained-run artifacts	84	84	Pass
Failed queue entries	0	0	Pass

The required campaign artifacts include the result CSV, gate summary, leaderboard, explicit best-run YAML and Markdown, artifact path list, campaign state, and baseline-topology comparison. Every trained run contains `best_model.pt` , `training_history.csv` , `test_predictions.npz` , and `metrics_summary.yaml` .

Branch Results

The error values below are MAE in `1e-3 deg` and are means across the three declared seeds for trained arms. Replay rows contain their single frozen result. Lower is better.

ID	Role	Runs	Gate outcome	Fw	Bw	Global
A00	Frozen global K01	1 replay	Control	1.401424	1.518885	1.460155
A00D	Routed directional K01	1 replay	Control	1.354961	1.579773	1.467367
A01	Decomposed global K01	1 replay	Identity pass	1.401424	1.518885	1.460155
A02	Forward centered H08	3	Pass, 3 of 3 seeds	1.397912	1.518885	1.458399
A03	H04 shape control	3	Specialty failed	1.399311	1.511419	1.455365
A04	F01 shape objective	3	Specialty and non-regression failed	1.396812	1.513841	1.455327
A05	S01 harmonic and closure	3	Non-regression failed	1.398555	1.518347	1.458451
A06	R00 dense condition library	3	Specialty and non-regression failed	1.403949	1.518515	1.461232
A07	S01 threshold control	3	Specialty failed	1.400321	1.518046	1.459184
A08	Passed branches only: A02	3	Executed; equivalent to A02	1.397912	1.518885	1.458399

Multi-Index Interpretation

Relative to the exact `A01` replay, `A02` provides a small forward raw-error gain while remaining exactly inactive on `Bw` . Its global test MAE improves from `0.001460155 deg` to `0.001458399 deg` , centered-shape MAE improves from `0.001220701 deg` to `0.001218450 deg` , and closure error improves from `0.000849561 deg` to `0.000820528 deg` . The offset surface is effectively unchanged. Its declared validation phase specialty and all required non-regression checks pass for every seed.

`A03` and `A04` show small test-set raw and shape gains, but neither passes its predeclared validation specialty gate. `A04` additionally fails the multi-index non-regression gate. `A05` improves closure and P95 behavior on test, but violates its validation non-regression contract. The dense and thresholded condition-library branches (`A06`)

and A07) do not qualify.

The routed directional K01 control (A00D) has the best forward replay MAE and lower global offset error than the global K01 topology, but worse backward MAE, global centered shape, and global P95. The campaign does not authorize a deployment-topology switch from this mixed evidence.

Provisional Scalar Winner

- Ablation: A04 .
- Seed: 271828 .
- Run: 2026-08-03-17-52-59__a04__seed_271828 .
- Parameters: 1,022 trainable specialist parameters.
- Best epoch: 39 .
- Global validation MAE: 0.001539071 deg .
- Global test MAE: 0.001456655 deg .
- Global test centered-shape MAE: 0.001214986 deg .
- Global test offset absolute error: 0.000577130 deg .
- Global test P95 curve MAE: 0.004183546 deg .
- Gate result: failed.

The scalar winner is not the accepted branch decision. Validation ordering selects the checkpoint, while the predeclared branch gate governs whether a specialist can advance. Only A02 advances to later official review.

Campaign Decision

The campaign decision is forward_harmonic_specialist_added at the empirical candidate level. The centered H08 contribution is admitted only behind its deterministic forward gate. It remains zero on Bw , excludes H08 a0 , and does not authorize the rejected global H08 formulation.

No second branch passed, so multi_specialist_candidate_qualified is false. Because A08 contains only A02 , it supplies no additional integration gain and should not be described as a distinct multi-specialist model.

Registry And Incumbent Decision

The new integrated_specialist_models family registry records the campaign's validation-only provisional winner and its failed-gate status. The program registry remains unchanged. K01 remains the cross-surface temporal offline leader, H08 remains the frozen forward-only non-temporal specialist, and the accepted periodic GRU and periodic harmonic MLP remain the deployment references.

No checkpoint is promoted into the curated accepted-model archive during this normal closeout.

Deployment Boundary

The campaign proves immutable artifact production, zero target-derived runtime inputs, deterministic direction gating, and bounded inspectable specialist heads in the Python workflow. It does not prove ONNX export for the new combined candidate, TwinCAT compilation, target activation, TF3820 licensing, ADS communication, PLC latency, or commissioned runtime compensation.

Manual PLC testing through the standalone TF3820 module remains a separate parallel activity.

Reproducibility Evidence

- Campaign root: `output/training_campaigns/2026-08-03-17-49-23_wave52r_integrated_specialist_model_2026_08_02` .
- Run root: `output/training_runs/integrated_specialist_models` .
- Result table: `campaign_results.csv` .
- Gate decision: `branch_gate_summary.yaml` .
- Scalar ordering: `campaign_leaderboard.yaml` .
- Provisional winner: `campaign_best_run.yaml` and `campaign_best_run.md` .
- Baseline topology evidence: `baseline_topology_comparison.yaml` .

Closeout Status And Next Step

Normal campaign closeout is complete after Markdown, PDF, registry, status, and active-state QA. The TE Program Status And Closeout Ledger was checked and updated because the campaign changed the integrated-specialist decision.

The heavy TE Curve Verification Pipeline remains a separate optional operator-run step. If approved, its launcher must support local and `-Remote` execution and must evaluate `Fw` , `Bw` , and `global` separately under the canonical multi-index curve-first policy. Until that later decision, A02 is a qualified empirical candidate only and A04 remains an unpromoted scalar winner.